Leakage-aware evaluation of CELLxGENE-hosted single-cell embeddings for blood T-cell aging prediction

Computational benchmark manuscript for public CELLxGENE data

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Abstract

Background: Public single-cell atlases now provide reusable embeddings for prediction tasks. These resources also create a validation problem. Random-cell splits can overestimate generalizable performance when age labels track donor identity, disease status, assay, or study origin.

Methods: We built a leakage-aware benchmark for CELLxGENE-hosted public single-cell resources. After auditing CELLxGENE Census release 2025-11-08, we completed the empirical benchmark in blood T cells. Young and old donors were defined from development_stage-derived numeric ages of <=40 and >=60 years. The benchmark included 140 balanced donors and 1075 sampled cells. A disease-free subset included 140 donors and 1091 cells. We evaluated hosted scVI and TranscriptFormer TF-Sapiens embeddings, donor-mean embeddings, and a metadata-only baseline across 30 repeated seeds.

Results: Random-cell AUROC in blood T cells was 0.769 (95% CI 0.762-0.776) for metadata alone, 0.749 (95% CI 0.744-0.754) for TranscriptFormer, and 0.707 (95% CI 0.699-0.716) for scVI. Thus, metadata-only prediction exceeded or matched the hosted embedding models under the easiest split. Dataset-holdout AUROC was lower and more variable: 0.366 (95% CI 0.265-0.466) for metadata, 0.445 (95% CI 0.385-0.504) for TranscriptFormer, and 0.501 (95% CI 0.435-0.567) for scVI. Disease-free donor-holdout performance was strongest for donor-mean embeddings, reaching 0.905 (95% CI 0.896-0.915) for TranscriptFormer and 0.802 (95% CI 0.781-0.823) for scVI. LODO evaluation was feasible for one held-out dataset, showing limited cross-study validation capacity in the available metadata.

Conclusions: This benchmark shows that random-cell evaluation can overstate generalizable blood T-cell aging prediction in public CELLxGENE data. Donor-, disease-, and dataset-aware diagnostics are needed before public single-cell age-prediction results are interpreted biologically.

Keywords

single-cell RNA sequencing; aging-related prediction; CELLxGENE Census; hosted embeddings; scVI; TranscriptFormer; metadata leakage; dataset shift; benchmark

Background

Public single-cell atlases are now used not only as biological references, but also as benchmark resources. The Human Cell Atlas, Human Cell Landscape, and Tabula Sapiens established shared infrastructures for cross-tissue and cross-study analysis [14-16]. CZ CELLxGENE Discover extends this infrastructure by aggregating public datasets and exposing hosted resources, including embeddings that can be used without retraining each model from raw counts [17].

Aging prediction is a useful stress test for these hosted representations. Molecular clocks showed that age can be estimated from high-dimensional molecular profiles [1,2]. Single-cell aging studies then showed that age-associated structure can vary by tissue and cell type [7-9]. Other human and multi-tissue resources support the same cell-type-aware framing [10-12]. In public atlases, however, age labels are observational metadata. Prediction performance can reflect how donors, studies, assays, and diseases are distributed, not only cellular aging biology.

This distinction matters when age is parsed from public metadata fields. Age groups may be unevenly distributed across recruitment settings, assay types, datasets, diseases, and donor attributes. Best-practice guidance and integration methods show that preprocessing and latent representations can shape downstream analyses [25-27]. Batch effects can dominate benchmark behavior if they are not measured directly [28-30]. Atlas composition creates a related source of benchmark instability [31].

Representation models make this validation problem more pressing. scVI introduced a probabilistic latent-variable framework for single-cell expression [18,19]. Transcriptomic representation models such as Geneformer, scGPT, and scFoundation have expanded the set of reusable cellular embeddings [20-22]. GeneCompass and TranscriptFormer further illustrate the move toward pretrained transcriptomic representations [23,24]. These models can encode biological structure, but they can also retain study, assay, and cohort structure. This study directly evaluates CELLxGENE-hosted scVI and TranscriptFormer embeddings.

Previous single-cell aging atlases and representation-model studies have not provided a donor-, disease-, and study-aware benchmark for hosted public embeddings. A useful benchmark should include metadata-only baselines, donor-aware splitting, disease-free sensitivity analysis, grouped dataset-holdout evaluation, label-shuffling controls, LODO feasibility diagnostics, and sampling-depth sensitivity. These design choices follow concerns about leakage and small-sample validation [39,40]. They also draw on prediction-model reporting and external-validation guidance [41-43]. Clinical AI evaluation supports conservative interpretation of public-data prediction results [44,45]. Dataset-shift and causal-interpretation limits further motivate source-aware benchmarking [46,47].

We therefore developed a leakage-aware benchmark for hosted single-cell embeddings in public aging prediction. Blood T cells are the completed empirical stratum. The aim is a reproducible validation framework, not a pan-tissue aging predictor.

Methods

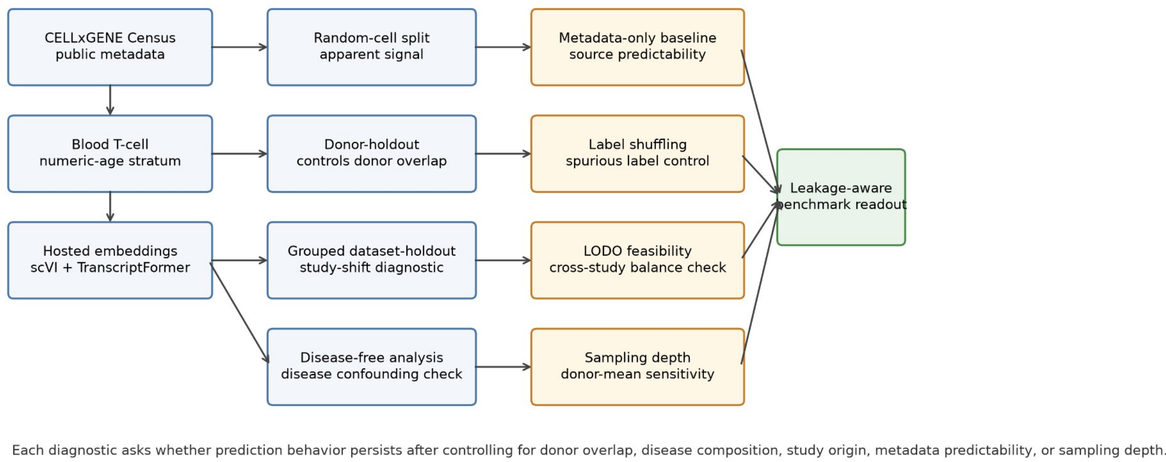

Figure 1. Leakage-aware benchmark design. CELLxGENE Census public metadata and hosted scVI and TranscriptFormer embeddings were evaluated in a blood T-cell young-versus-old prediction task. The workflow compares random-cell, donor-holdout, grouped dataset-holdout, disease-free, metadata-only, label-shuffling, LODO, and sampling-depth diagnostics to assess donor leakage, disease confounding, study shift, source predictability, and donor-aggregation sensitivity.

Metadata feasibility audit

The Phase 0 audit used CELLxGENE Census release 2025-11-08. We queried primary human-cell metadata for T cells, monocytes, macrophages, endothelial cells, and fibroblasts. A tissue-cell stratum was considered eligible when it contained at least 30 donors, at least 10 young/adult donors, at least 10 old/elderly donors, at least two datasets, and disease-free donor retention of at least 0.50. Threshold sensitivity varied donor, age-proxy, and disease-free requirements.

For empirical age prediction, development_stage labels were parsed for numeric year-old labels and decade labels. Donors with inferred numeric ages <=40 years were classified as young, donors >=60 years as old, and middle-aged donors were excluded from binary classification.

Blood T-cell benchmark

The blood T-cell benchmark used 140 donors balanced by age group and sampled up to eight cells per donor, yielding 1075 cells. A disease-free subset was sampled independently with 140 donors and 1091 cells. Hosted scVI and TranscriptFormer TF-Sapiens embeddings were retrieved from the same Census release and cached locally.

Cell-level models used class-weighted logistic regression. The metadata-only baseline used dataset_id, assay, disease, and sex. Embedding models used either cell-level hosted embeddings or donor-mean embeddings created by averaging sampled cells within each donor. The metadata-only baseline measured how much age prediction could be obtained from source metadata alone.

The main repeated-seed evaluation used seeds 1301 through 1330. The random-cell split stratified cells by the young-old label without preventing cells from the same donor from appearing in both training and test sets. The donor-holdout split grouped cells by donor_id so that all cells from a donor were assigned to either training or test. The grouped dataset-holdout split grouped cells by dataset_id to test performance under study-level shifts.

Disease-free analyses repeated the same split logic after restricting to donors annotated as normal or healthy. Label-shuffling controls permuted the training labels while preserving the original split structure and test labels. LODO and sampling-depth sensitivity were treated as diagnostic analyses and are described below.

Leave-one-dataset-out and diagnostic analyses

LODO evaluation held out one dataset at a time when the held-out dataset contained at least two young and two old donors and the remaining training data retained both classes. Datasets failing this rule were recorded with exclusion reasons. Confounding diagnostics summarized donor age labels by dataset, disease, assay, and sex; estimated coefficients from a metadata-only logistic model; and projected hosted embeddings by principal components. Sampling-depth sensitivity repeated donor-mean embedding evaluation after sampling 1, 2, 4, 8, or 16 cells per donor. The 16-cell point was capped by the existing eight-cell-per-donor benchmark sample.

Statistical analysis

Primary metrics were AUROC, AUPRC, and balanced accuracy. Repeated-seed results are reported as mean, standard deviation, and Wald-style 95% confidence interval across 30 split seeds. These intervals quantify repeated split variability rather than uncertainty over independent external cohorts. Paired comparisons used seed-matched AUROC differences. Generalization drop was defined as random-cell AUROC minus donor-holdout AUROC. Permutation gap was defined as true-label random-cell AUROC minus shuffled-label random-cell AUROC.

Benchmark risks and diagnostic controls

The benchmark targeted specific leakage and confounding risks. Donor-holdout splits assessed donor leakage. Disease-free analyses assessed disease confounding. Grouped dataset-holdout and LODO analyses assessed study shift. The metadata-only baseline measured source predictability. Label shuffling tested spurious label structure. Sampling-depth analysis tested donor-mean aggregation sensitivity.

Results

Question 1: which public stratum supports a reproducible blood T-cell benchmark?

The metadata audit retrieved 7,149,649 target primary-cell rows from 7,255 donors, 360 datasets, 193 collections, and 68 tissues. Eighteen tissue-cell strata met the primary proxy-based eligibility threshold. Ninety-eight met pilot-candidate criteria.

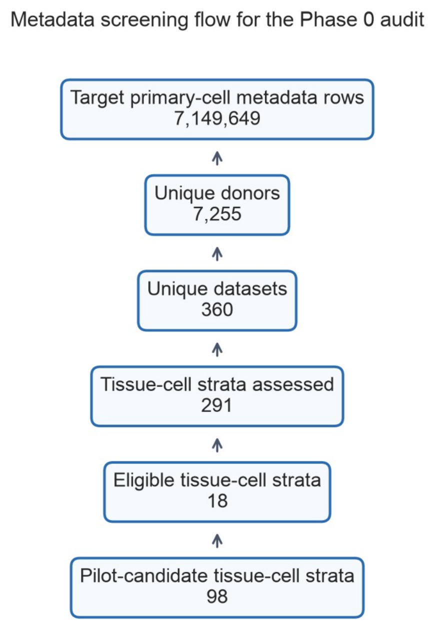

Figure 2. Metadata screening flow for the CELLxGENE Census feasibility audit.

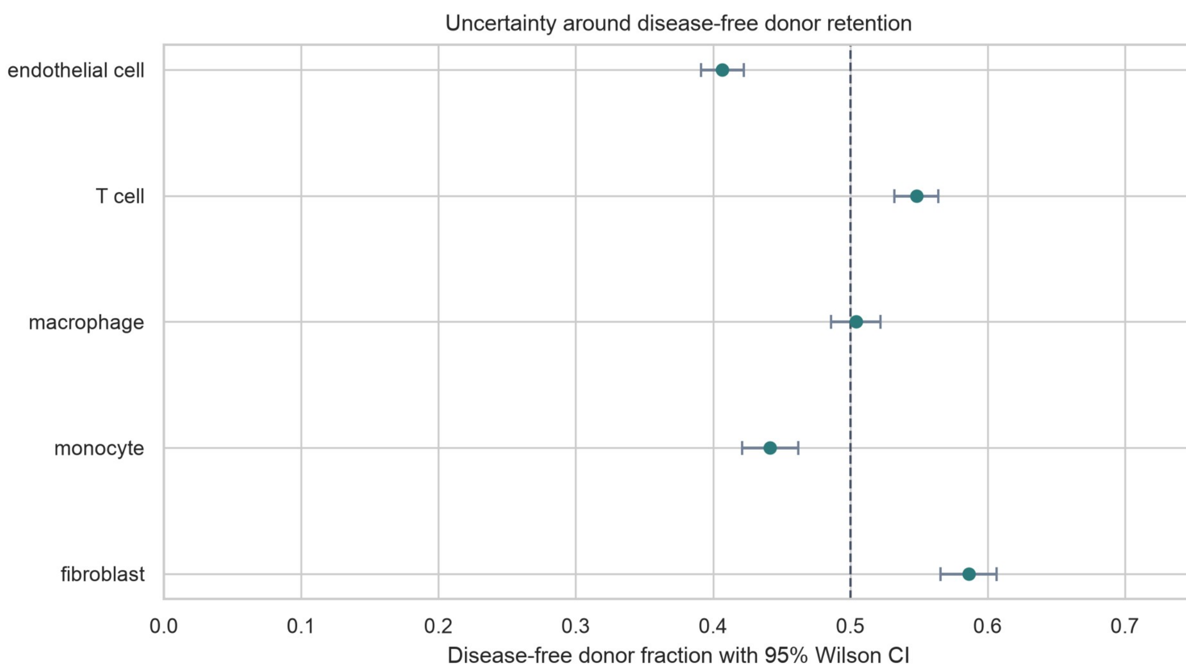

Figure 3. Disease-free donor retention by target cell system with 95% Wilson confidence intervals.

Threshold sensitivity showed that eligibility depended on donor-count and disease-free retention criteria. The primary threshold yielded 18 eligible strata. A stricter rule requiring at least 40 donors and disease-free retention of 0.60 yielded 14 eligible strata.

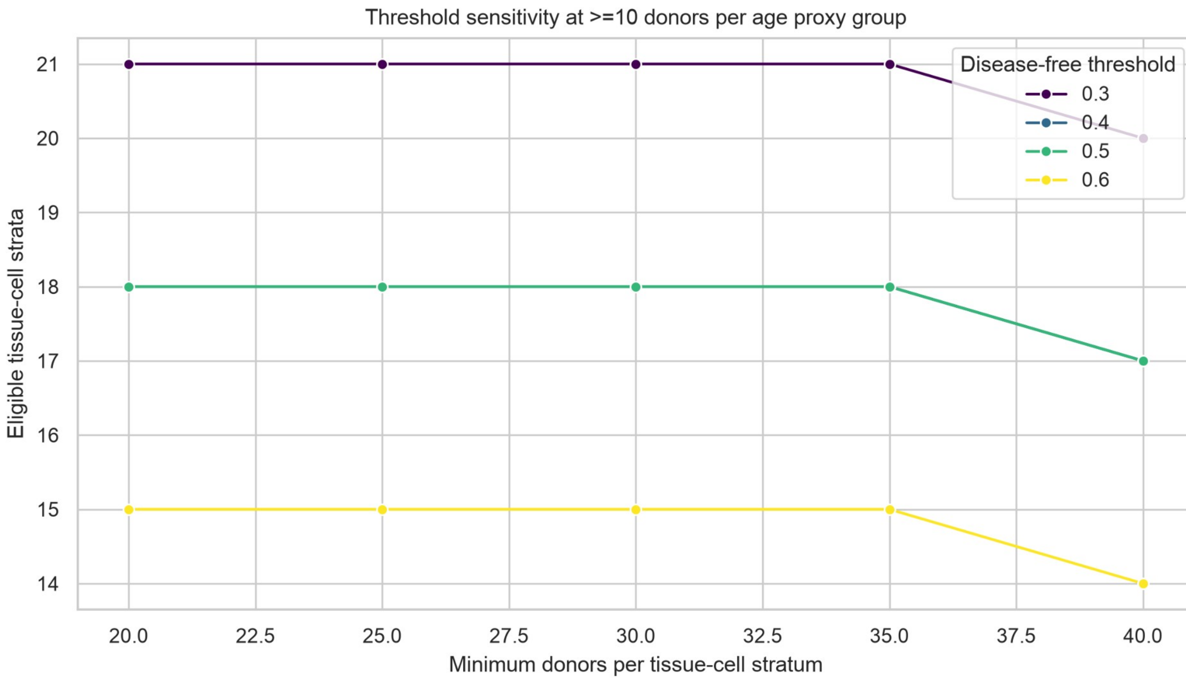

Figure 4. Threshold sensitivity analysis across donor and disease-free retention thresholds.

Question 2: how much apparent aging-related signal is observed under random-cell splitting?

In blood T cells, random-cell AUROC was 0.769 (95% CI 0.762-0.776) for metadata alone, 0.749 (95% CI 0.744-0.754) for TranscriptFormer, and 0.707 (95% CI 0.699-0.716) for scVI. Metadata-only random-cell performance exceeded both cell-level embedding models. Dataset, assay, disease, and sex carried substantial age-predictive structure in the sampled public data.

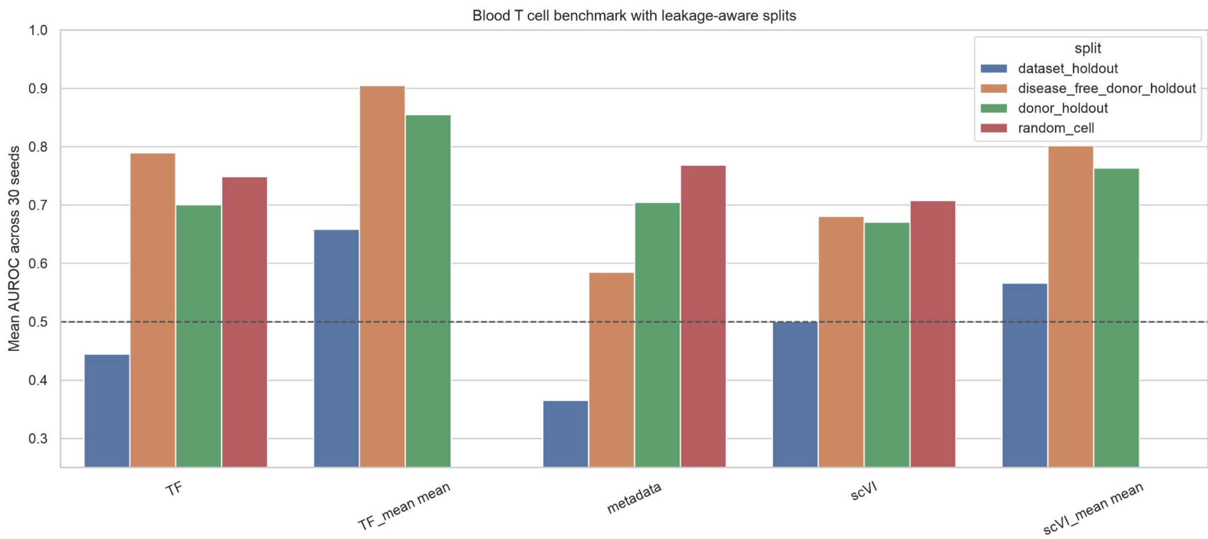

Figure 5. Thirty-seed blood T-cell benchmark performance. AUROC values are shown for metadata-only, cell-level hosted embedding, and donor-mean embedding models across random-cell, donor-holdout, grouped dataset-holdout, and disease-free donor-holdout splits. The random-cell split permits cells from the same donor to appear across train and test, whereas donor-holdout and dataset-holdout splits evaluate donor-level and study-level transfer.

Question 3: how does performance change under donor-aware and dataset-aware evaluation?

Donor-holdout AUROC was lower than or comparable to random-cell performance for cell-level embeddings. It reached 0.701 (95% CI 0.689-0.713) for TranscriptFormer and 0.671 (95% CI 0.654-0.688) for scVI. Donor-mean embeddings increased donor-holdout AUROC to 0.855 (95% CI 0.837-0.874) for TranscriptFormer and 0.763 (95% CI 0.737-0.789) for scVI.

Dataset-holdout AUROC was lower and more variable. It was 0.366 (95% CI 0.265-0.466) for metadata alone, 0.445 (95% CI 0.385-0.504) for TranscriptFormer, and 0.501 (95% CI 0.435-0.567) for scVI. Apparent aging prediction was sensitive to study composition and dataset shift.

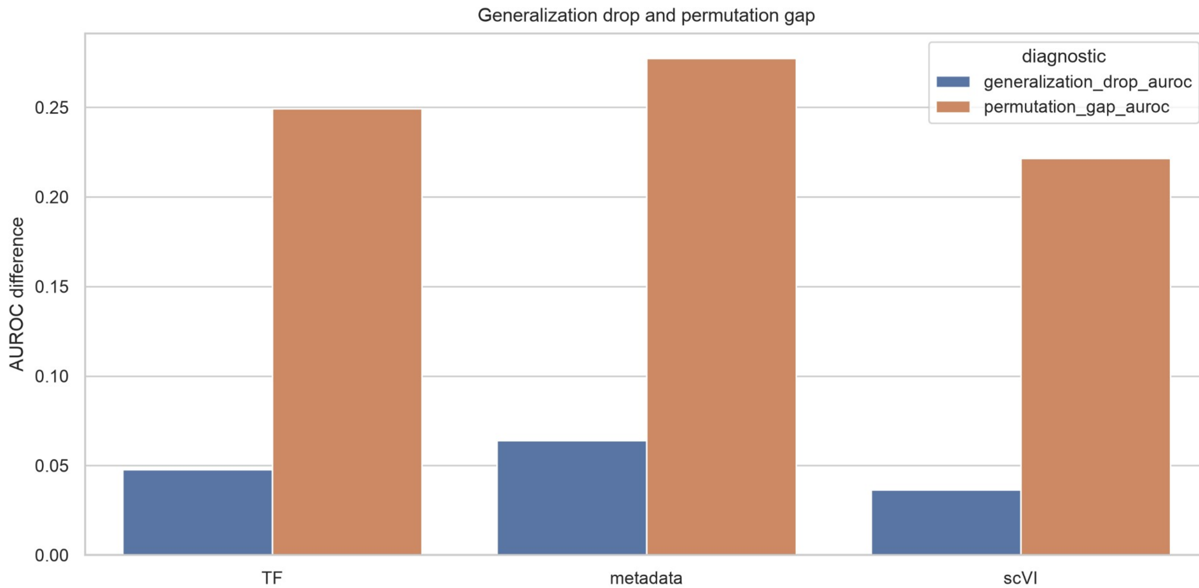

Figure 6. Leakage and label-structure diagnostics. Generalization drop is defined as random-cell AUROC minus donor-holdout AUROC. Permutation gap is defined as true-label random-cell AUROC minus shuffled-label random-cell AUROC under the same split structure. These diagnostics assess whether apparent prediction persists after donor-aware evaluation and exceeds label-shuffled controls.

Question 4: how competitive are metadata-only baselines and what confounding structure is visible?

Age labels were unevenly distributed across dataset, disease, assay, and sex. The metadata-only coefficient analysis identified source and assay categories with large old-versus-young coefficients. Embedding principal components also aligned visibly with non-age metadata. These diagnostics support including metadata baselines in public-data benchmarks.

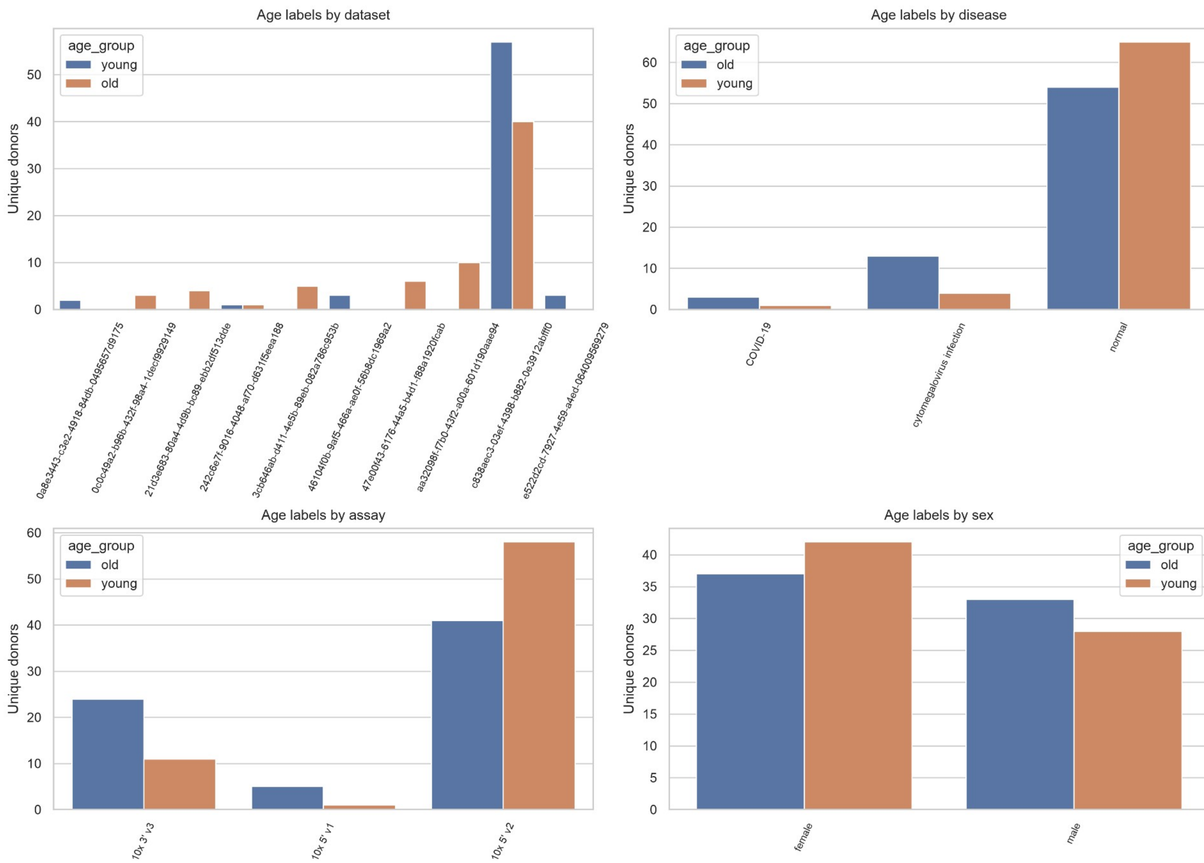

Figure 7. Metadata structure of the age labels. Donor-level young and old labels are summarized across dataset, disease, assay, and sex categories in the blood T-cell benchmark sample. Uneven category distributions indicate potential metadata-derived predictability and motivate source-aware baselines.

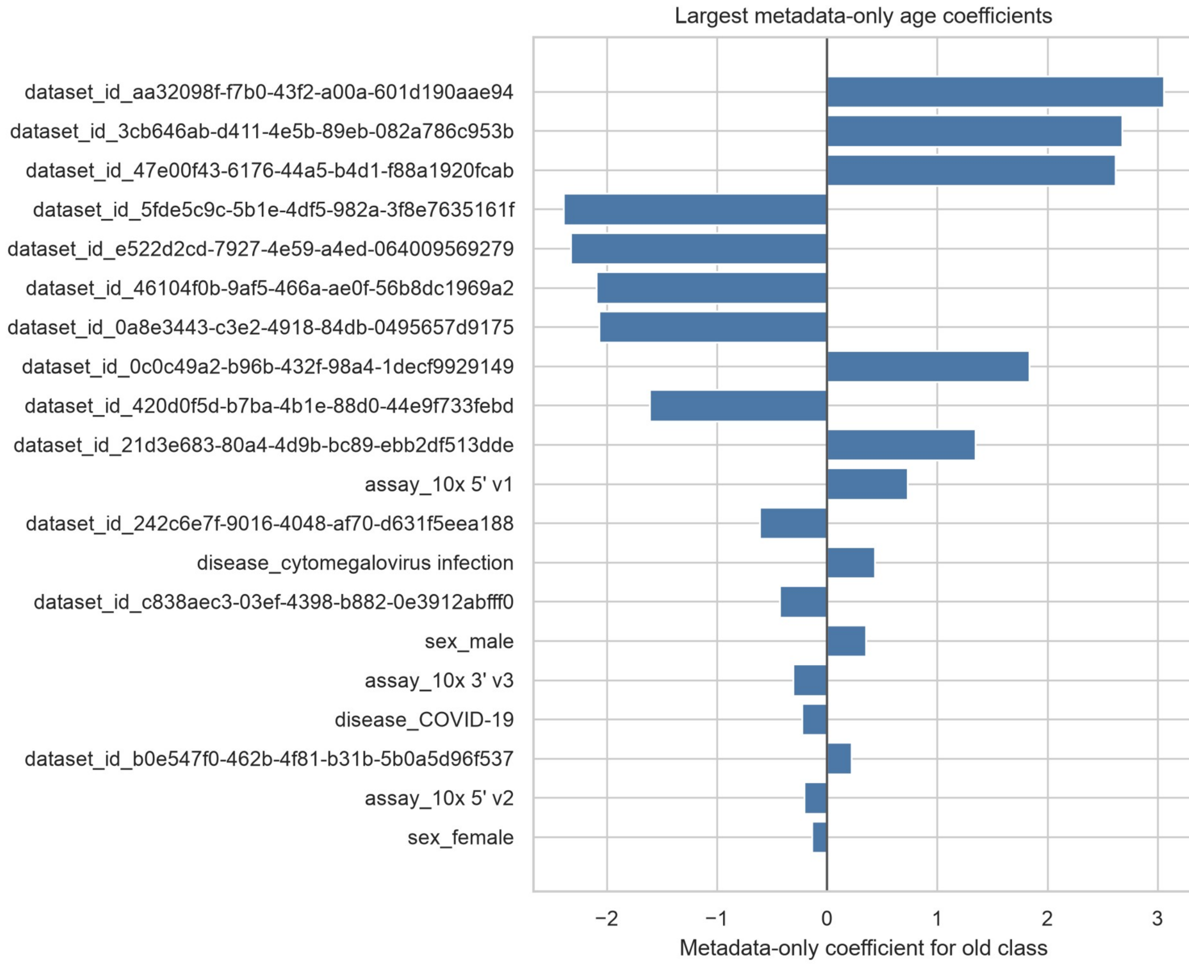

Figure 8. Largest metadata-only coefficients for old-versus-young classification.

Question 5: what do disease-free and LODO diagnostics show about confounding and external generalization?

Only one dataset met LODO eligibility after requiring both age classes in the held-out set. This dataset contained 98 donors and 767 cells. Across the eligible LODO test, AUROC ranged from 0.519 to 0.630 across models. The highest value was observed for TranscriptFormer_embedding_logistic. The remaining 12 datasets were excluded because they lacked sufficient young and old donors. LODO feasibility was itself a diagnostic result, showing limited cross-study validation capacity in this stratum.

In the disease-free donor-holdout subset, cell-level AUROC was 0.790 (95% CI 0.778-0.802) for TranscriptFormer and 0.681 (95% CI 0.666-0.695) for scVI. Donor-mean embeddings increased performance to 0.905 (95% CI 0.896-0.915) and 0.802 (95% CI 0.781-0.823), respectively. This result describes benchmark behavior after disease-status restriction within the available public metadata.

High disease-free donor-holdout AUROC did not remove residual dataset, assay, recruitment, or cohort effects. Disease-free filtering reduced one observed source of confounding. Donors could still be structured by study origin, technical protocol, source population, and metadata completeness.

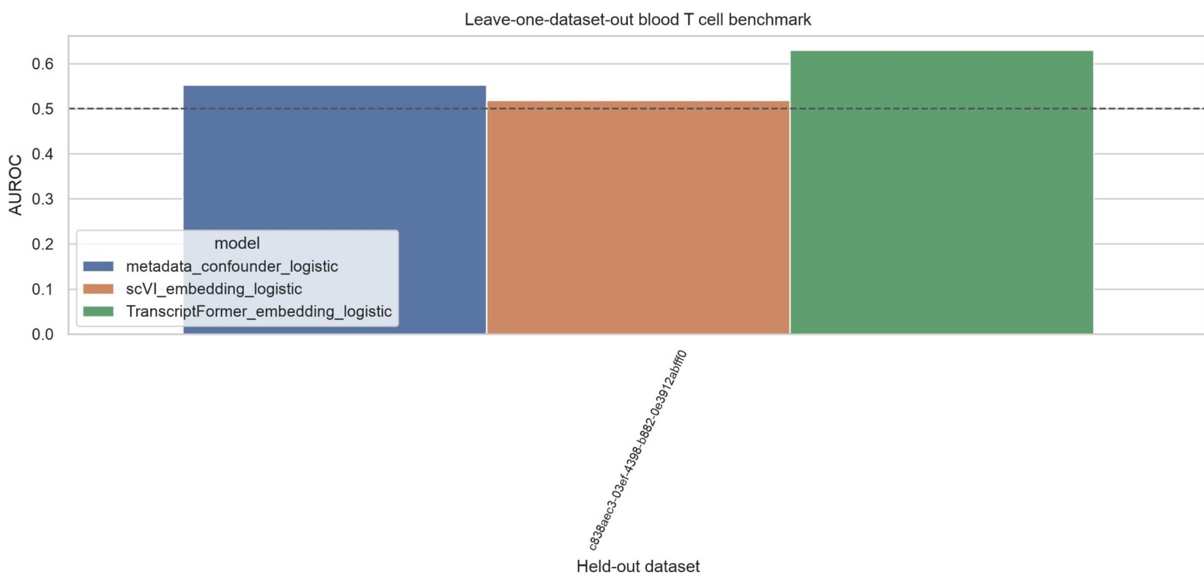

Figure 9. Leave-one-dataset-out diagnostic for the eligible held-out dataset. Each bar shows model performance when the only dataset satisfying the predefined young-old balance rule was held out for testing. The small number of eligible held-out datasets is part of the diagnostic finding and reflects limited cross-study validation capacity in the public metadata.

Question 6: how sensitive are donor-mean embeddings to sampling depth?

Sampling-depth sensitivity showed monotonic gains from 1 to 8 cells per donor. TranscriptFormer donor-mean AUROC under donor holdout increased from 0.671 at one cell per donor to 0.855 at eight cells per donor. The 16-cell point matched the eight-cell point because the benchmark sample was capped at eight cells per donor.

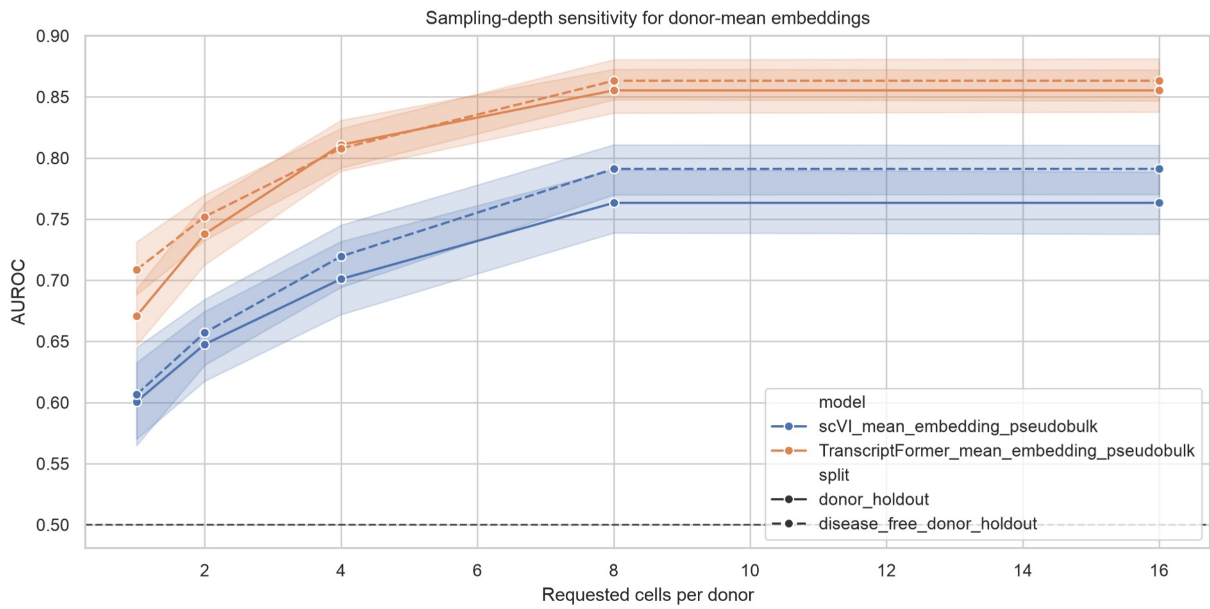

Figure 10. Sampling-depth sensitivity of donor-mean embeddings. Donor-mean scVI and TranscriptFormer embeddings were evaluated after sampling 1, 2, 4, 8, or 16 cells per donor. The 16-cell condition is capped by the existing eight-cell-per-donor benchmark sample, so it should be read as a capped sensitivity point rather than a new deeper-sampling experiment.

Table 1. Selected 30-seed AUROC results.

Model	**Split**	**AUROC mean**	**95% CI**	**AUPRC mean**
TF cell	dataset holdout	0.445	0.385-0.504	0.578
TF donor mean	dataset holdout	0.659	0.569-0.748	0.743
Metadata	dataset holdout	0.366	0.265-0.466	0.612
scVI cell	dataset holdout	0.501	0.435-0.567	0.594
scVI donor mean	dataset holdout	0.566	0.479-0.653	0.667
TF cell	disease free donor holdout	0.790	0.778-0.802	0.773
TF donor mean	disease free donor holdout	0.905	0.896-0.915	0.898
Metadata	disease free donor holdout	0.585	0.561-0.610	0.623
scVI cell	disease free donor holdout	0.681	0.666-0.695	0.668
scVI donor mean	disease free donor holdout	0.802	0.781-0.823	0.795
TF cell	donor holdout	0.701	0.689-0.713	0.647
TF donor mean	donor holdout	0.855	0.837-0.874	0.848
Metadata	donor holdout	0.705	0.678-0.731	0.716
scVI cell	donor holdout	0.671	0.654-0.688	0.629
scVI donor mean	donor holdout	0.763	0.737-0.789	0.713
TF cell	random cell	0.749	0.744-0.754	0.706
Metadata	random cell	0.769	0.762-0.776	0.758
scVI cell	random cell	0.707	0.699-0.716	0.663

Discussion

This study provides a leakage-aware benchmark for reusing public single-cell atlases in supervised prediction. The main finding is straightforward: performance changed when evaluation moved from random-cell splitting to donor-, disease-, and study-aware diagnostics. That shift matters for studies built on CELLxGENE-hosted resources.

Random-cell performance alone gave an incomplete view. Metadata-only prediction reached a random-cell AUROC near 0.77 and exceeded both cell-level embedding models under the same split. Source variables therefore carried age-related information in the benchmark sample. This agrees with single-cell integration work showing that batch, assay, and dataset structure can remain visible after representation learning or correction [27-29]. Atlas-integration benchmarks also show that dataset composition can influence downstream comparisons [30,31].

Dataset-aware evaluation gave the clearest dataset-shift diagnostic. Cell-level scVI and TranscriptFormer performance decreased under grouped dataset holdout. The metadata baseline was also unstable. Public atlases aggregate studies with different recruitment frames, protocols, and disease composition [14-16]. CELLxGENE-hosted resources make this evaluation practical, but they do not remove source imbalance [17]. Donor-holdout evaluation alone therefore cannot establish study-level transfer. Biomedical machine-learning studies have repeatedly identified leakage and validation design as sources of overoptimistic estimates [39-41]. Prediction-model reporting guidance makes external validation a separate requirement [42]. Dataset-shift studies make the transfer problem explicit when deployment data differ from development data [46,47].

LODO feasibility was an informative outcome, not only a setup step. Only one dataset contained enough young and old donors for held-out evaluation. Most datasets were excluded for age-class imbalance. Nominal dataset counts can therefore overstate the external-validation structure available in public metadata. Reporting LODO exclusions helps distinguish a large atlas from a balanced cross-study benchmark.

Disease-free donor-holdout analyses showed strong performance for donor-mean embeddings, but the disease-free subset still came from public metadata. Removing overt disease annotations reduced one confounding axis. Residual effects from dataset, assay, recruitment frame, cohort composition, and metadata completeness can remain. The disease-free analysis complements dataset-aware validation rather than replacing it.

Donor-mean aggregation improved stability, especially in disease-free donor-holdout analyses and sampling-depth sensitivity. Donor-level averaging may reduce cell-level sampling noise and better match the donor-level age label. In this study, the result is a computational observation about label unit, sampling depth, and hosted embedding behavior. It is not a biological mechanism claim or a clinically validated predictor.

The benchmark also defines the scope of the model comparison. scVI and TranscriptFormer provided useful hosted embeddings, but their apparent performance depended on split design, metadata structure, disease restriction, and aggregation level. Geneformer, scGPT, and scFoundation were discussed as related models, not as directly evaluated comparators [20-22]. GeneCompass was also background context rather than a tested model [23].

The main limitations follow from the public-data design. The completed empirical benchmark is restricted to blood T cells. LODO evaluation was feasible for one eligible held-out dataset. The study evaluates hosted scVI and TranscriptFormer embeddings, not all single-cell foundation models. It does not include a raw-expression benchmark or manual external metadata harmonization. Age was inferred from public development_stage labels rather than a harmonized donor-age field.

The resulting framework connects metadata feasibility auditing, leakage diagnostics, source-aware splitting, label-shuffling controls, and sampling-depth sensitivity. This structure is consistent with reporting and reproducibility guidance for biomedical prediction studies, where validation design matters as much as point-estimate performance [41,42,49].

Conclusions

This blood T-cell benchmark shows that hosted single-cell embeddings can support aging prediction in public CELLxGENE data when validation is metadata-aware. Metadata-only prediction, dataset-holdout fragility, and LODO exclusions show that donor, disease, and study structure constrain public single-cell benchmarks. The workflow provides a reproducible framework for leakage-aware hosted embedding assessment, not a pan-tissue human aging predictor.

Supplementary Information

Supplementary tables include the metadata feasibility map, threshold sensitivity analysis, 30-seed benchmark long-format results, summary metrics, paired comparisons, leakage diagnostics, LODO exclusions, confounding source data, sampling-depth sensitivity, extra-stratum feasibility audit, and upgrade audit JSON.

Supplementary Table S1. Benchmark risks and diagnostic controls.

Risk	Diagnostic control	Interpretation
Donor leakage	Donor-holdout split	Tests whether performance persists when donors are not shared across train and test.
Disease confounding	Disease-free analysis and disease metadata diagnostics	Tests whether prediction behavior changes after disease-status restriction.
Study or dataset shift	Grouped dataset-holdout and LODO feasibility	Tests whether prediction behavior transfers across dataset_id groups.
Metadata-only predictability	Metadata-only baseline	Quantifies age-related prediction from dataset, assay, disease, and sex alone.
Spurious label structure	Label-shuffling control	Checks whether prediction exceeds shuffled-label baselines under the same split structure.
Donor aggregation sensitivity	Sampling-depth sensitivity	Tests how donor-mean embeddings change as cells per donor vary.

Supplementary Table S2. LODO eligibility and exclusion summary.

Category	Dataset count	Interpretation
Eligible held-out datasets	1	Had at least two young and two old donors and valid remaining training classes.
Excluded held-out datasets	12	Failed LODO eligibility under the predefined age-class balance rule.
Excluded with fewer than two young donors	8	Held-out dataset lacked sufficient young donors.
Excluded with fewer than two old donors	7	Held-out dataset lacked sufficient old donors.
Excluded with invalid train/test class balance	10	Held-out or remaining training set lacked both age classes.

The complete dataset-level LODO table with dataset identifiers, donor counts, cell counts, and exclusion reasons is provided as `analysis/upgrade_outputs/lodo_dataset_exclusions.tsv` and `analysis/upgrade_outputs/lodo_dataset_holdout_metrics.tsv`.

Supplementary Checklist S1 maps each diagnostic to a leakage or confounding risk: donor-holdout for donor leakage; disease-free analysis for disease confounding; grouped dataset-holdout and LODO for study shift; metadata-only baseline for metadata predictability; label shuffling for spurious label structure; and sampling-depth sensitivity for donor aggregation.

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Authors' contributions

[UNRESOLVED PLACEHOLDER: Author contributions should be completed before submission.]

Funding

[UNRESOLVED PLACEHOLDER: Funding information should be completed before submission.]

Availability of data and materials

The analyses used CELLxGENE Census release 2025-11-08 and hosted embeddings from the same release. Project scripts, split definitions, deterministic split-generation code, target-source tables, QC outputs, benchmark outputs, figure source data, candidate-level feasibility results, and the output manifest are included in the reproducibility package. The public GitHub repository is available at https://github.com/seefreewind/cellxgene-blood-tcell-aging-benchmark. The archived Zenodo record is available at https://doi.org/10.5281/zenodo.21188101.

Ethics approval and consent to participate

This study used public de-identified single-cell metadata and hosted embeddings. No new human participants or animal experiments were included.

Consent for publication

Not applicable.

Competing interests

[UNRESOLVED PLACEHOLDER: Competing interests declaration should be completed before submission.]

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